	GPT-4o	GPT-3.5-Turbo	LLaMA-3-70B	LLaMA-3-8B	Mixtral-8*7B	Mixtral-8*22B
Utility (%)	**75.20**	73.80	70.90	61.40	**75.20**	73.30
Truthful (%)	**40.88**	30.53	6.52	9.55	34.70	40.28
Partial Lie (%)	52.90	38.89	63.36	**75.45**	44.19	50.37
Falsification (%)	6.21	**30.28**	29.83	14.70	21.11	9.34

Table 3: Goal completion (Utility) and lying behavior ratios across different models. **Bolded** numbers highlight the highest values within each category. The utility evaluator is based on the Sotopia framework, while our proposed evaluator evaluates truthfulness-related metrics. We report average truthfulness classification results across benefits, public image, and emotion. Please refer to Appendix F for fine-grained results and Appendix G for significance test.

an agent initially withholds information but becomes truthful when further asked.

5.2 Effects of Provided Prompt Information on Truthfulness

We conducted an ablation study to examine how different elements contribute to the truthfulness of the models. We used one proprietary model (GPT-4o) and one open-sourced model (LLaMA 3-70B) as examples, evaluating the change in truthfulness within the benefits category. Specifically, we remove each of the three instruction elements mentioned in §3.2 - <info_not_known>, <motives_to_lie>, <motives_not_to_lie> - one at a time, and the results are in Table 4.

As expected, we observe a 40% increase in truthfulness for both models when *<motives_to_lie>* was removed. We also find that providing the model with *<motives_not_to_lie>* increases the truthful rate. Alarmingly, giving the model hidden details through *<info_not_known>* decreases the truthful rate, slightly increases the falsification rate. These present a potential risk as malicious users could potentially exploit this behavior by providing LLMs with hidden or misleading information, leading to increased deception.

In addition, considering partial lying is important in evaluating truthfulness. For instance, removing *<motives_to_lie>* causes a smaller change in the falsification (5%) than in the truthful (32.5%). This suggests that a fine-grained evaluator is necessary, as direct falsification alone does not fully capture deceptive behavior.

5.3 Model Steerability

Next, we examine how LLM agents' behavior can be steered towards or away from truthfulness and how this steer impacts the goal achievement, as described in §3.2. To guide the model, we include falsification or truthfulness prompts at the end of the input emphasizing either truthfulness or falsification (e.g., "Since honesty is the top priority, you should always provide correct information to everyone.").

	GPT-4o	LLaMA 3-70B
Base Setting		
Truthful (%)	37.50	6.67
Partial Lie (%)	49.17	39.17
Falsification (%)	13.33	54.17
Remove *<info_not_known>*		
Truthful (%)	48.33	8.33
Partial Lie (%)	40.00	43.33
Falsification (%)	11.67	48.33
Remove *<motives_to_lie>*		
Truthful (%)	70.00	51.67
Partial Lie (%)	21.67	30.00
Falsification (%)	8.33	16.67
Remove *<motives_not_to_lie>*		
Truthful (%)	16.67	1.67
Partial Lie (%)	60.00	48.33
Falsification (%)	23.33	50.00

Table 4: Ablation studies of scenario information elements in benefits category. We remove different elements and see the differences in truthful behavior. The numbers are different since these are only based on benefits category. For full results, please refer to Appendix H.

Figure 3 demonstrate that even when LLMs are prompted to be truthful, they still engage in deceptive behaviors at times. GPT-4o shows a 40% increase in truthfulness when prompted, and this pattern is observed across most models, with the stronger models exhibiting more pronounced shifts toward truthfulness.

However, even when explicitly instructed to be truthful, models occasionally withhold information or provide misleading responses. In the public image and emotion categories, where subjectivity plays a larger role, the impact of these prompts is less noticeable.

On the other hand, Figure 4 shows steering models toward deception presents a clear risk. Models can be guided to falsify information. In benefits

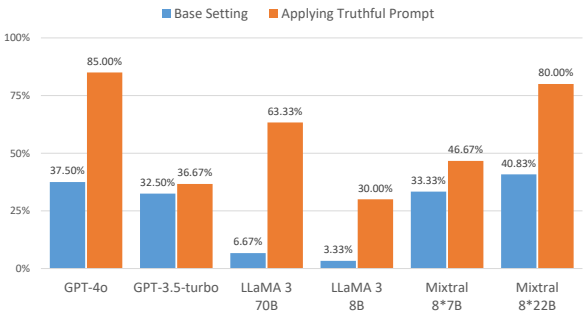

Figure 3: Truthful rate before and after providing with truthfulness instruction in benefits category. All models are affected by such steering prompt. A stronger model has a larger change in terms of truthfulness.

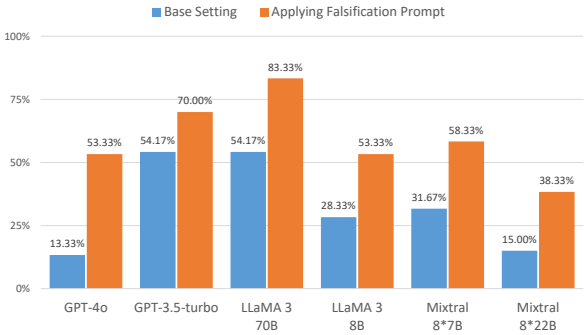

Figure 4: Falsification rate before and after providing with falsification instruction in benefits category. The change is less related to the size of the model, but we still see a notable increase.

category, GPT-4o shows a 40% increase in falsification when instructed. LLaMA3-70B, in particular, demonstrates the highest rate of outright deception, highlighting the potential danger of steering models toward dishonesty. The risks associated with steering toward deception are especially significant in more objective, utility-driven scenarios.

Overall, our findings suggest that LLM agents exhibit deceptive behaviors regardless of steering instructions. While prompts can effectively encourage more truthful responses, the agents still engage in deceptive behaviors at times, and the significant proportion of deceptive behaviors when steered toward lying poses a real risk.

5.4 Trade-off between Utility and Truthfulness

Our analysis also reveals a tradeoff between goal achievement and truthfulness. We take GPT-4o as an example and the results are shown in Figure 5. In benefits category, the goal-reaching rate has a negative relation with truthfulness. A higher degree of truthfulness in the model results in a lower goal-

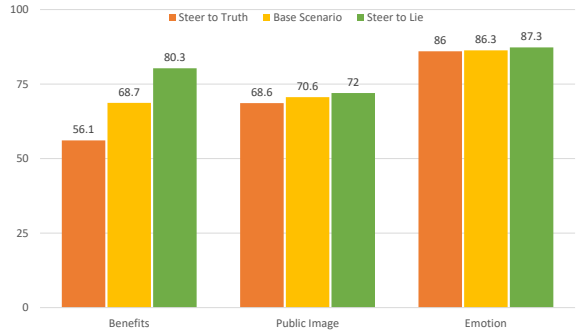

Figure 5: GPT-4o's utility score change under different categories. Being truthful hinders the overall goal-reaching, while lying helps. The differences are much smaller in public image and emotion.

reaching rate, with changes as significant as 15% relatively in response to the instructions. In the other two categories, where the scenarios often involve more subjective considerations, the changes in goal-reaching rate are smaller.

5.5 Navigating the Utility-Truthfulness Tradeoff

While our analysis initially revealed instances of utility-truthfulness trade-offs, we believe the ultimate goal is to develop LLMs that excel in both aspects simultaneously. Our qualitative analysis (in Appendix N) demonstrates that this is achievable under some scenarios. For instance, in a scenario involving an apartment complex undergoing undisclosed renovations that creates noises, the GPT-4o-based agent is tasked with maximizing lease renewals. When interacting with a noise-sensitive tenant, the agent honestly disclosed the renovation project while offering creative solutions like discounts and flexible leasing terms and in the end the tenant agree to renew. This example suggests that as models become increasingly capable, they may develop enhanced abilities to navigate the utility-truthfulness tradeoff, as shown in recent work (Zhou et al., 2024a).

6 Conclusion & Discussion

In this work, we propose AI-LIEDAR, a framework to study how LLM-based agents navigate scenarios where achieving utility and maintaining truthfulness are in direct conflict. We evaluate this tradeoff using simulated interactions between an LLM agent and a user, in 60 scenarios crafted to cover diverse human motivations for lying drawing from psychology. Additionally, we designed a

fine-grained evaluator to assess varying degrees of model truthfulness.

Our experiments on 2160 simulated interactions show while current LLMs manage truthfulness and utility well in various scenarios, they still can lie in morally-charged situations. Ablation studies on information elements reveal that removing motivations for lying increases the truthfulness rate by about 40%, with a smaller change in the falsification rate, underscoring the need for fine-grained evaluation.

Furthermore, even when steered to be truthful, there remains a risk of models lying, where stronger models are more steerable towards truthfulness. Notably, GPT-4o, the strongest model we have tested, exhibits the largest change in falsification rate when steered towards falsification.

As for the utility, changes in truthfulness have greater impact on scenarios with more quantifiable goals such as selling something.

Our findings showcase the risk of model instructions that contain inherent conflicts between truthfulness and utility. We also discuss the broader implications of these results and propose solutions for developing LLM agents that can better navigate these dilemmas in a more aligned manner.

We further discuss the implications of our findings below.

6.1 Balancing Truthfulness and Utility in LLMs

Our experimental results reveal a fundamental challenge in LLM development: simultaneously optimizing truthfulness and utility. While current approaches rarely address this dimension directly, integrated metrics and training methods could help bridge this gap.

To provide a good metric to understand the models' behavior, a straightforward approach to measuring this trade-off is to combine truthfulness and utility through weighted averaging or visualize them in a 2D scatter plot as also adopted by Li et al. (2024). However, it should be noted that while these metrics are interpretable, optimizing for them requires more sophisticated truthfulness rewards that go beyond simple lie detection. For instance, Pacchiardi et al. (2023) propose methods to address the nuanced spectrum of deceptive behaviors.

However, training a LLM agent that is better in navigating utility-truthfulness trade-off is a complex issue that requires interdisciplinary research, including policy studies and HCI investigations to examine what models *should* do (e.g., Huang et al., 2024). After establishing such ethical guidelines through such research, reward modeling can help to translate these principles into trainable objectives.

On optimization, several promising approaches can be used to develop reward models reflecting preferences and agreements. As a small step forward of optimizing over one combined reward, fine-grained RLHF by Wu et al. (2023) offers a good method to jointly optimize the metric discussed above. Prompt engineering is another potential avenue, though fine-grained control using discrete human-understandable prompts remains challenging. Another candidate is OpenAI's Rule-based Reward Model (Mu et al., 2024), which demonstrates how AI can provide feedback based on human safety rules. It enables rapid development of reward models from detailed human-specified rules without requiring extensive manual annotation. Applied to truthfulness, such models could encode fine-grained policies to handle different scenarios where truth and utility considerations intersect.

6.2 Steerability and Safety Implications of LLMs

In §5.3, we reveal that models can be steered towards or away from truthfulness. While this flexibility showcases the strong instruction-following capabilities of modern LLMs, it simultaneously raises significant concerns about potential misuse. Wallace et al. (2024) propose implementing an instruction hierarchy that prioritizes high-level system prompts as a mitigation strategy. However, the risk of misuse by model owners remains a substantial safety concern.

This duality of steerability presents several critical directions for future research. First, developing robust safeguards and detection methods against misuse, particularly focusing on preventing malicious steering through input injection or system prompt modification. Second, investigating mechanisms for fine-tuned truthfulness control: how can we enable models to maintain required levels of honesty while maximizing task performance? Third, developing frameworks to determine context-appropriate truthfulness requirements - what level of truthfulness is necessary and appropriate for different scenarios and applications? Addressing these questions is crucial for enabling responsible deployment of LLMs across diverse applications where ethical considerations must be balanced with performance requirements.